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Graph-based Two-Three Wheeler Classification in Unconstrained Indian Roads

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***Abstract*— Driven by the continuous advancements in the field of Advanced Driver Assistance Systems (ADAS), object detection and classification are crucial functionalities receiving significant attention in recent research. Mainly, Indian scenarios are highly dynamic due to the diverse and challenging driving conditions, including congested urban traffic, unpredictable road infrastructure, and varying weather patterns. This paper presents a unique method for classifying two and three-wheelers on the challenging Fine-Grained Vehicle Detection (FGVD) dataset from India. In the first stage, we employ the yolov7-tiny model to detect the objects in the image. In the second stage, we represent the detected object images in the graph domain and utilize a GNN architecture to classify objects within the image. A Spatial Graph Convolutional Network (SGCN) is employed in the second stage to classify the detected objects based on their spatial context. This approach leverages the strengths of SGCNs for classification, potentially leading to improved performance on the two, and three-wheelers of the FGVD dataset. We evaluate our method on the FGVD dataset and demonstrate its effectiveness in achieving object detection and classification.**

I. INTRODUCTION

Advanced Driver-Assistance Systems (ADAS) are experiencing a significant transformation due to technological advances. The incorporation of various sensor technologies and robust algorithms is shaping the future of ADAS [1]. Researchers and industries constantly seek innovative solutions to real-world challenges such as adverse weather, low-light conditions, and dynamic traffic patterns to ensure safety, efficiency, and adaptability in diverse driving scenarios. However, driving scenarios' dynamic and unstructured nature poses a particular hurdle for ADAS functionality. In India, where unpredictable traffic patterns, congested roads, and diverse driving behaviors are prevalent, effective ADAS implementation is a significant challenge [2]. Accurate object detection and classification are crucial for robust perception and decision-making in autonomous vehicles. This paper proposes a novel approach that utilizes Graph Neural Networks (GNNs) to address these challenges and improve object classification performance in Indian traffic conditions.

Traditional object detection and classification methods in ADAS focus on pixel-wise information, which can struggle in complex traffic scenes with occlusions or unstructured layouts. GNNs offer an approach by modeling the spatial relationships between objects in an image as a graph [3]. This allows them to capture contextual information and interactions between objects, leading to potentially more robust object classification, especially in dynamic environments like Indian roads. Despite the advancements in GNNs for various tasks, their application in Indian ADAS scenarios remains largely unexplored. Most of the existing research has primarily focused on conventional methods [4], and this gap presents a significant opportunity to leverage the power of GNNs for improved perception in Indian traffic conditions. This paper aims to propose a novel GNN-based approach for object classification on the Fine-Grained Vehicle Detection (FGVD) dataset, paving the way for further research in this exciting area. SGCN helps explore spatial connections within the image, which is crucial for object classification tasks; by representing these connections as graphs, SGCN can effectively leverage spatial information to enhance performance.

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A. Literature Work

Object detection and classification are fundamental tasks for Advanced Driver-Assistance Systems (ADAS) to perceive and understand the surrounding environment. Traditional methods for object detection rely on techniques like hand-crafted features and feature engineering, followed by classifiers like Support Vector Machines (SVMs) or decision trees [5], [6]. However, these methods struggle with complex scenes and variations in object appearance [7]. Recent advancements in deep learning have revolutionized the field of object detection and classification for ADAS. Convolutional Neural Networks (CNNs) have emerged as the dominant approach, with architectures like YOLO [8], Faster R-CNN [9], and SSD [10] achieving state-of-the-art performance on benchmark datasets. These methods learn effective feature representations directly from the data, leading to improved robustness and generalization. Considerable progress has been made in object detection and classification for ADAS using convolutional neural networks, resulting in impressive results [11], [12], [13], [14]. On the other hand, despite limited initial research works [2], [15], research on object detection and classification for ADAS in Indian driving scenarios is an under-explored area with significant potential for future advancements. Additionally, combining information from cameras, LiDAR, and radar, sensor fusion techniques is gaining traction to enhance object detection and classification performance in challenging environmental conditions [16], [17]. However, graph neural networks have not been extensively explored in this field yet.

Many researchers working on autonomous mobility rely on publicly available datasets like ADE20K [18], CamVid [19], Mapillary Vistas [20], KITTI [21], Cityscapes [22], and nuScenes [23], and BDD100K [24]. However, these datasets mainly contain driving scenarios from regions with

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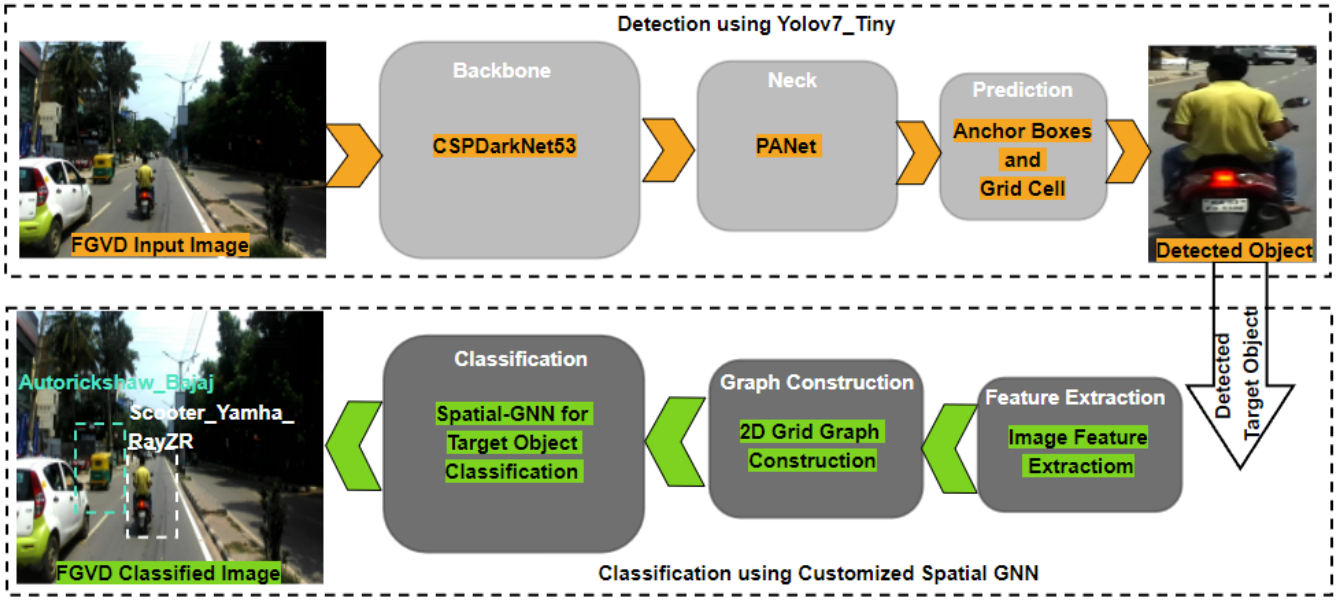

Fig. 1. Overview diagram illustrating the workflow of the proposed method for object detection and classification. The diagram showcases implementing the YOLOv7-tiny model for object detection, followed by extracting image features and their representation in a 2D grid graph. The SGCN then utilizes graph representation to classify detected objects.

well-defined regulations and standardized traffic patterns, such as European countries. Although some datasets include North America and Singapore recordings, they often lack the rich diversity of traffic conditions encountered in India. The FGVD dataset [25] is unique in that it offers imagery from Indian roads, characterized by factors like dense traffic, diverse vehicle types (including auto-rickshaws), and unpredictable driving behaviors. This characteristic makes FGVD a valuable resource for developing robust ADAS functionalities tailored to Indian traffic scenarios.

B. Contributions

In the first stage of our process, we consider the YOLOv7-tiny model for object detection. The YOLOv7-tiny model utilizes a lightweight architecture for real-time object detection on resource-constrained devices. In the second stage, we classify the detected objects in the FGVD dataset. We first resize the images for object classification and then extract the image features. Then, we represent the images in the graph domain using the 2D grid graph. Within this graph, each pixel functions as a node, and the features extracted from the photos serve as attributes for these nodes. In our experiments, we use three types of image features: the RGB feature, the texture feature, and the gradient-based features. These 2D grid graphs act as input to the SGCN network, classifying the detected objects. Fig. 1 elucidates our two-stage approach for object detection and classification in the FGVD dataset. On the other hand, Two- and three-wheeler detection is crucial for robust ADAS in Indian traffic scenarios due to their prevalence and unpredictable driving patterns. These smaller vehicles often weave through traffic, making them difficult to anticipate and posing a significant safety risk. Accurate detection and classification of two- and three-wheelers is essential for autonomous vehicles to navigate Indian roads safely and navigate the dynamic interactions between different vehicle types. That is why we do more experiments on object detection and classification, specifically for the two- and three-wheelers. The major contributions of our work are as follows:

- We utilize the YOLOv7-tiny model for object detection in the FGVD dataset and SGCN for object classification.
- We experiment with various image features as node attributes in the graph representation of the images.
- We perform extensive experiments, specifically for the detection and classification of two- and three-wheelers in the FGVD dataset.

II. FGVD: Fine-Grained Vehicle Detection

Unlike many popular datasets focusing on standardized traffic conditions, FGVD offers imagery captured on Indian roads. This data incorporates the complexities of Indian traffic, including dense traffic flow, diverse vehicle types (like two, and three-wheelers), and unpredictable driving behaviors. These characteristics make FGVD a well-suited platform for evaluating and refining object detection and classification algorithms for ADAS in the Indian context.

A. Data Acquisition

The Indian Driving Dataset (IDD) [2] captures the complexities of navigating roadways in India. Data acquisition involved driving along various routes in Bangalore and Hyderabad, encompassing both urban and rural areas. This resulted in data collection that reflect diverse driving conditions, including highways, single and double-lane roads, and unstructured environments with extensive construction zones. The FGVD dataset considers 5502 out of 16311 images from

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the IDD Detection dataset based on the size of vehicle boxes, occlusion, and traffic density.

B. Dataset Features

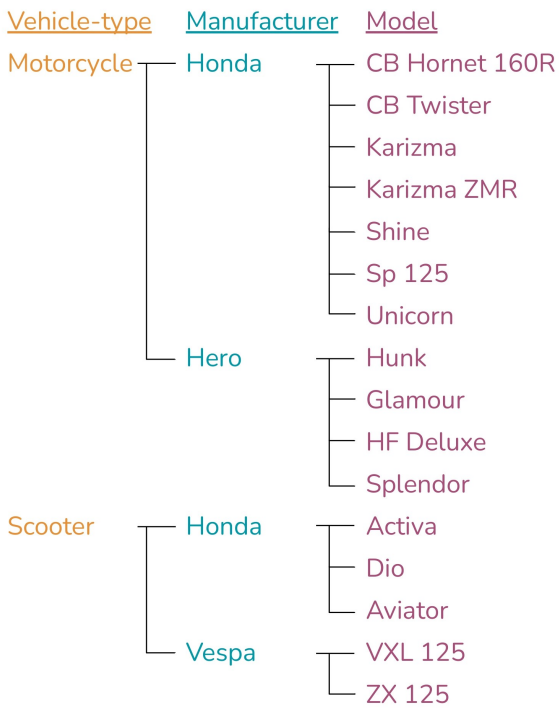

Fig. 2. Label Hierarchy Illustration for Two-wheelers [25]

The FGVD dataset comprises 5502 scene images annotated with approximately 24,450 bounding boxes. These bounding boxes are associated with fine-grained labels (217 total, 210 unique labels, and seven repeated labels from higher hierarchical levels). Notably, the dataset considers the variability in vehicle sizes by employing different threshold ratios for bounding boxes. For example, the threshold for trucks is higher than that for cars, and the threshold for cars is higher than for bikes. Additionally, the hierarchical labeling scheme in FGVD varies by vehicle type, as illustrated in Fig. 2 and 3, two-wheelers have a three-level hierarchy, whereas three-wheelers have a two-level hierarchy.

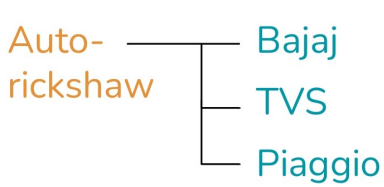

Fig. 3. Label Hierarchy Illustration for Three-wheelers [25]

C. Statistical Composition

Fig. 4 depicts the distribution of vehicles within the FGVD dataset, and Table I presents the statistics of labels assigned to each vehicle type.

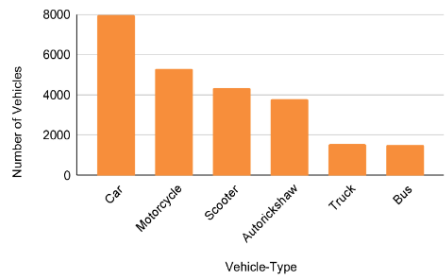

Fig. 4. Distribution of Vehicles within the FGVD Dataset [25]

TABLE I
LEVELS OF HIERARCHY FOR EACH VEHICLE TYPE IN FGVD [25]

Vehicle Type (L-1)	Levels of Hierarchy	L-2 Labels	L-3 Labels
Car	3	22	112
Motorcycle	3	11	67
Scooter	3	9	23
Truck	2	7	7
Autorickshaw	2	6	6
Bus	2	2	2
Total	3	57	217

III. METHODOLOGY

Our proposed methodology is divided into two sections. The first section addresses object detection within the image, while the second section focuses on classifying the detected objects.

A. Object Detection using Yolov7-tiny Model

Object detection using the YOLOv7-tiny model [26] enables real-time identification and localization of multiple objects within an image. YOLOv7-tiny offers a trade-off between speed and accuracy, making it well-suited for applications requiring real-time object detection on resource-constrained devices. The YOLOv7-tiny model architecture consists of a convolutional neural network (CNN) backbone, followed by detection layers responsible for predicting bounding boxes and object classes. This model typically utilizes a lightweight CNN backbone, such as CSPDarknet, to extract features from the input image. These features are then passed to the detection head. The detection head comprises convolutional layers that predict bounding boxes and associated object probabilities. In YOLOv7-tiny, this head is optimized for speed and efficiency, making it suitable for real-time applications. Each bounding box is represented by coordinates (x, y) of its centre, width (w), height (h), and confidence score, indicating the likelihood of containing an object used for bounding box prediction. Along with bounding boxes, the YOLOv7-tiny model predicts the probability distribution over predefined object classes for each bounding box. Non-maximum suppression (NMS), a post-processing step, is applied to eliminate redundant detections. NMS removes bounding boxes with low confidence scores and suppresses overlapping bounding boxes, retaining only the most confident and non-overlapping.

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B. Object Classification using SGCN

In the second stage, we resize the images to ensure consistent input size. Next, we extract features from the resized images. The features considered are RGB features, texture, and gradient-based features. We then represent the images in the graph domain using a 2D grid graph. In this graph, each pixel in the image corresponds to a node, and the extracted features become the attributes associated with each node. Finally, we feed these 2D grid graphs as input to SGCN for object classification.

1) Image Feature Extraction: We basically use three types of features:

- RGB feature: The RGB feature is one of the image features we utilize as node attributes in the 2D grid graph. This feature captures the basic colour information of each pixel in the image. We connect the graph nodes with edges with the Euclidean distance between pixel intensities as weights. This allows the SGCN to learn how the overall object appearance is constructed from the interplay of different pixel regions within the image, aiding in object classification.
- Texture Information: We incorporate texture information into the 2D grid graph using Gabor filters [27]. These filters, defined mathematically as:

$$G(x, y; \lambda, \theta, \psi, \gamma) = \exp\left(-\frac{x^2 + \gamma^2 y^2}{2\sigma^2}\right) \cos\left(2\pi\left(\frac{x}{\lambda}\cos(\theta) + \frac{y}{\lambda}\sin(\theta)\right) - \psi\right) \tag{1}$$

  where λ represents wavelength, θ denotes orientation, ψ is phase offset, and γ controls aspect ratio, are applied to the image to capture local variations in texture. The resulting Gabor features are then used as node attributes in the graph.
- Gradient-based feature: For the gradient-based feature we consider the Sobel operator [28]. The Sobel operator is a well-known edge detection technique that highlights image regions with significant intensity changes. Mathematically, it utilizes two 3×3 kernels, one for horizontal edges (h_x) and one for vertical edges (h_y), defined as:

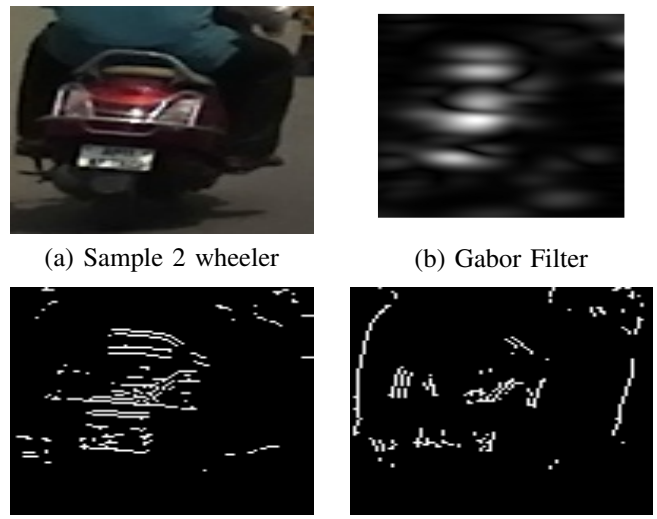
$$h_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad h_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

  These kernels are applied as convolutions to the image, resulting in two separate maps representing the strength of horizontal and vertical edges at each pixel location. The Sobel magnitude, often used for edge detection, can be calculated as:

$$|G| = \sqrt{G_x^2 + G_y^2} \tag{2}$$

  where G_x and G_y are the results of the convolutions with h_x and h_y kernels, respectively. Including the Sobel magnitude as node attribute might be particularly helpful for tasks like vehicle type classification where edges often play a role in defining object boundaries and shapes.

As illustrated in Fig. 5, the proposed method extracts texture and gradient-based features for a sample-detected two-wheeler image.



(a) Sample 2 wheeler (b) Gabor Filter

(b) Sobel Horizontal (b) Sobel Vertical

Fig. 5. Sample 2 Wheeler Image and the Applied Image Features which act as Node Attributes

2) Graph Construction: In our approach, we represent the images as 2D grid graphs. In this representation, each pixel in the image corresponds to a node in the graph, and the connections between nodes are established based on their spatial proximity within the grid. This allows the graph to capture not only the inherent features extracted from each pixel (like RGB or texture information) but also the spatial relationships between these features. This approach has shown promise in recent image classification tasks. The authors in [29], [30] use the 2d grid graphs for image classification using GNNs.

3) Application of SGCN: In our approach, we leverage SGCNs [31] to exploit the power of the 2D grid graph representation for object classification. Unlike traditional GCNs that focus solely on spectral properties of the graph [32], SGCNs incorporate spatial information to learn more effective feature representations. The core operation in an

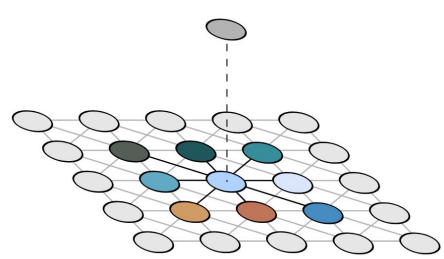


Fig. 6. Spatial Convolutional Filter used in SGCN [31]

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SGCN layer is the spatial convolutional filter (depcited in Fig. 6), defined mathematically as:

$$\mathbf{H}' = \hat{\mathbf{D}}^{-\frac{1}{2}}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}\mathbf{W} + \mathbf{b} \tag{3}$$

where:

- $\mathbf{H}$ represents the feature matrix of the previous layer, with each row corresponding to a node and each column representing a feature dimension. For the input layer, the feature matrix will consist of the node feature vectors, which are the image features of each pixel.
- $\mathbf{H}'$ denotes the updated feature matrix after the SGCN layer.
- $\hat{\mathbf{A}}$ represents the normalized adjacency matrix of the 2D grid graph, incorporating both node connections and their spatial proximity.
- $\hat{\mathbf{D}}$ is the diagonal degree matrix of $\hat{\mathbf{A}}$, where each element on the diagonal corresponds to the sum of inbound weights for a specific node's connections.
- $\mathbf{W}$ is the weight matrix of the SGCN layer, to be learned during the training process.
- $\mathbf{b}$ is the bias vector.

The key difference from standard GCNs [33] lies in the normalized adjacency matrix $\hat{\mathbf{A}}$. It incorporates a weighting scheme that considers not only whether two nodes are connected but also their spatial distance within the grid. This allows the SGCN to learn feature representations that take into account the relative positions of pixels in the image, leading to a more comprehensive understanding of the object's structure. By stacking multiple SGCN layers, the network progressively refines the feature representations, ultimately enabling accurate object classification.

IV. Results and Discussion

We conduct experiments using the FGVD dataset consisting of 3535 images for training, 884 images for validation, and 1083 images for testing. Following detection, the object images are resized to a uniform resolution of 64×64 pixels before feature extraction commences. In our study, we explore four distinct cases to evaluate the performance of our approach. The first case involves classifying two-wheelers versus all other vehicles, where vehicles other than two-wheelers are grouped as 'other_vehicles'. Similarly, the second case focuses on distinguishing three-wheelers from other vehicles. The third case compares two-wheelers and three-wheelers against all other vehicle classes. Finally, the fourth case encompasses the classification of all vehicle types. We train the Spatial Graph Convolutional Networks (SGCNs) following the architecture described in [31]. For the graph convolutional layers, we utilize 64 filters with a filter size of 3×3 and ReLU activation functions. We employ a learning rate of 0.001 with the Adam optimizer and a batch size of 32. Additionally, we apply a dropout rate of 0.5 to prevent overfitting during training. The Gabor filter we employ utilizes a wavelength of 6 pixels, a frequency of 1.0, and a gamma of 1.0 for texture feature extraction. As evidenced in Tables II, III, and IV, the combination of RGB and Gabor filters as node attributes achieves the best performance for L-2 and L-3 labels, while the combination of RGB, Gabor filters, and Sobel operator leads to superior results for L-1 classification. This pattern is also observed for separate experiments conducted on two- and three-wheeler datasets.

To evaluate the effectiveness of our approach, we compare its performance with the YOLOv5+HRN model, as presented in [25]. The analysis of this comparative study is presented in Table V. In terms of time complexity, SGCN generally has a lower complexity compared to the HRN model. SGCN simplifies the graph convolution operation, reducing the

TABLE II
Results for Vehicle Type Classification (L-1) across Different Feature Types and Cases

Feature Type	Two-wheelers vs All	Three-wheelers vs All	Two-wheelers vs Three-wheelers vs All	All
RGB Features	90.27%	87.27%	84.16%	84.62%
Gabor Filter	79.28%	74.55%	70.47%	61.71%
Sobel Operator	92.14%	90.28%	81.95%	82.12%
RGB+Gabor	92.24%	89.12%	82.52%	80.41%
RGB+Sobel	93.44%	90.49%	84.18%	83.73%
RGB+Gabor+Sobel	**95.05%**	**93.24%**	**88.15%**	**86.37%**

TABLE III
Results for L-2 across Different Features and Cases

Feature Type	Two-wheelers vs All	Three-wheelers vs All	Two-wheelers vs Three-wheelers vs All	All
RGB Features	47.89%	45.23%	43.56%	41.78%
Gabor Filter	58.34%	55.67%	53.21%	51.45%
Sobel Operator	38.24%	36.78%	34.56%	31.89%
RGB+Gabor	**78.46%**	**76.23%**	**75.12%**	**71.89%**
RGB+Sobel	49.98%	47.34%	44.76%	42.12%
RGB+Gabor+Sobel	69.79%	67.46%	65.21%	63.07%

TABLE IV
Results for L-3 across Different Feature

Feature Type	Two-wheelers vs Car
RGB Features	43.56%
Gabor Filter	41.78%
Sobel Operator	19.21%
RGB+Gabor	**49.87%**
RGB+Sobel	36.45%
RGB+Gabor+Sobel	47.23%

TABLE V
Results for comparison with other models

Model	L-1 label	L-2 label	L-3 label
YOLOv5+HRN	83.21%	59.02%	48.40%
RGB+Gabor	80.41%	**71.89%**	**49.87%**
RGB+Gabor+Sobel	**86.37%**	63.07%	47.23%

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number of parameters and computational steps. This leads to faster training and inference times.

V. Conclusion

In this work, we investigate the application of graph convolutional networks (GCNs) to object classification within the domain of Indian ADAS. The first stage involves Object detection using the yolov7-tiny model. The second stage concerns detected object image resizing, feature extraction, and object classification. In the second stage, we represent the images as 2D grid graphs, where each pixel corresponds to a node and features become node attributes. This representation is fed into a Spatial Graph Convolutional Network (SGCN) for object classification. This work paves the way for further exploration in this domain. Future directions include incorporating domain-specific knowledge into the SGCN learning process. Overall, this research highlights the promising potential of GCNs for object classification in Indian ADAS.

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